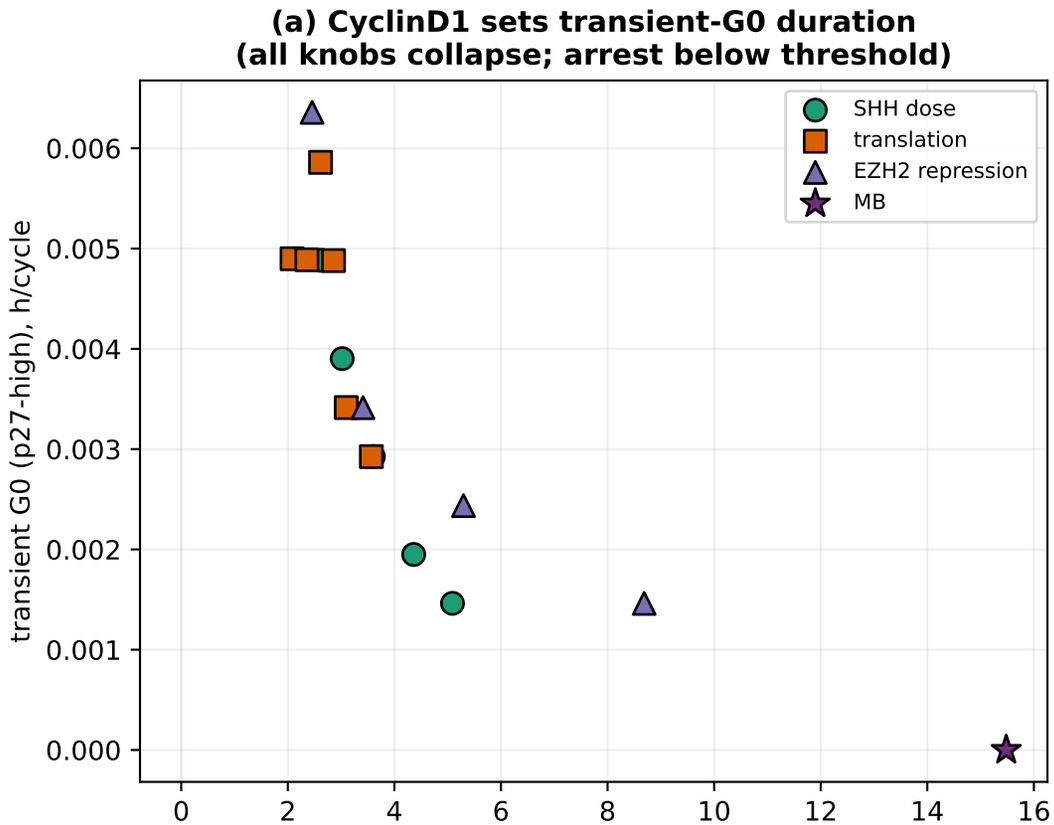
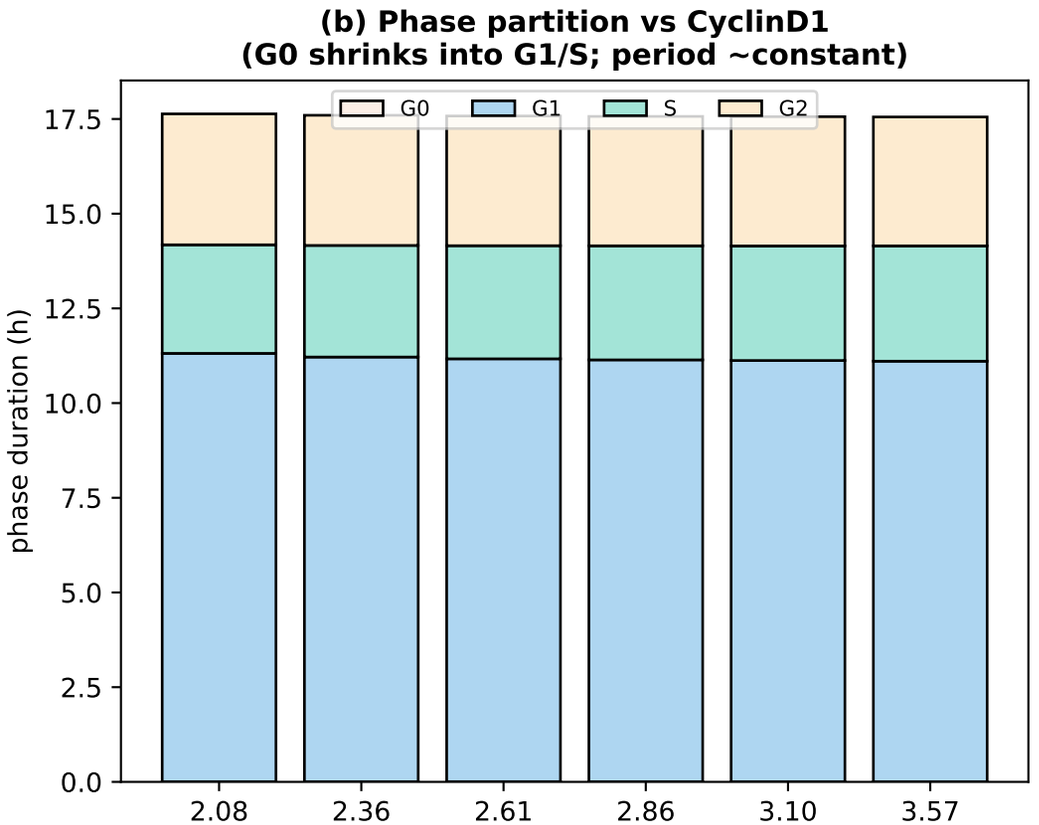
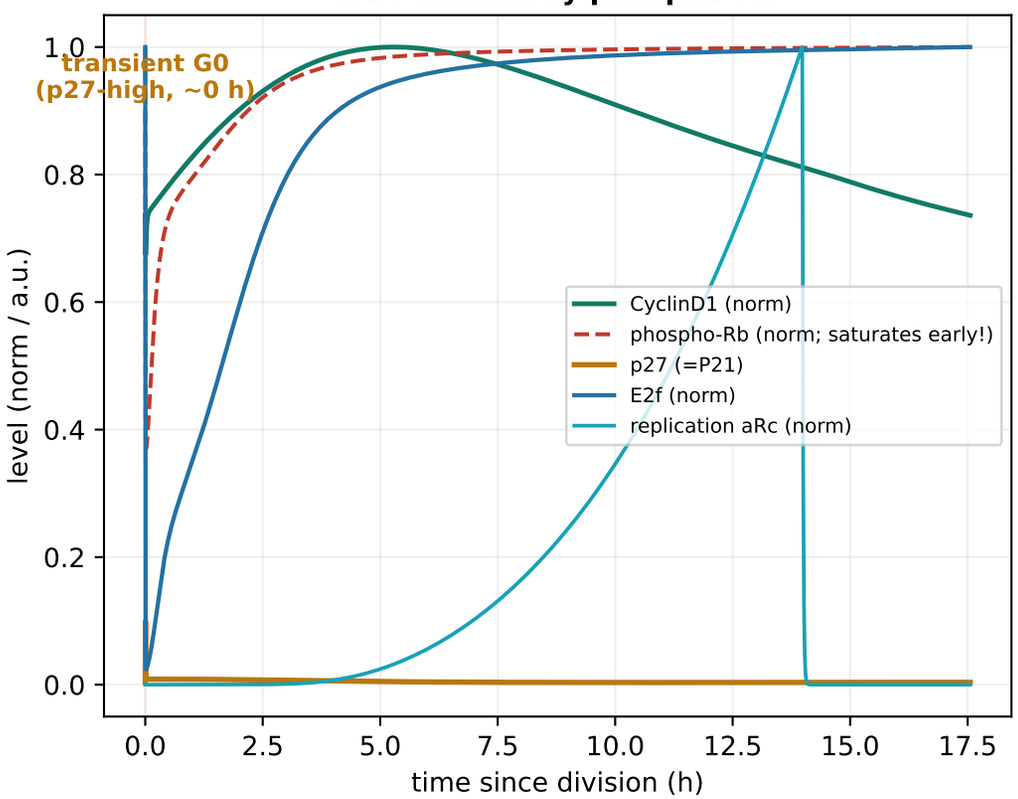
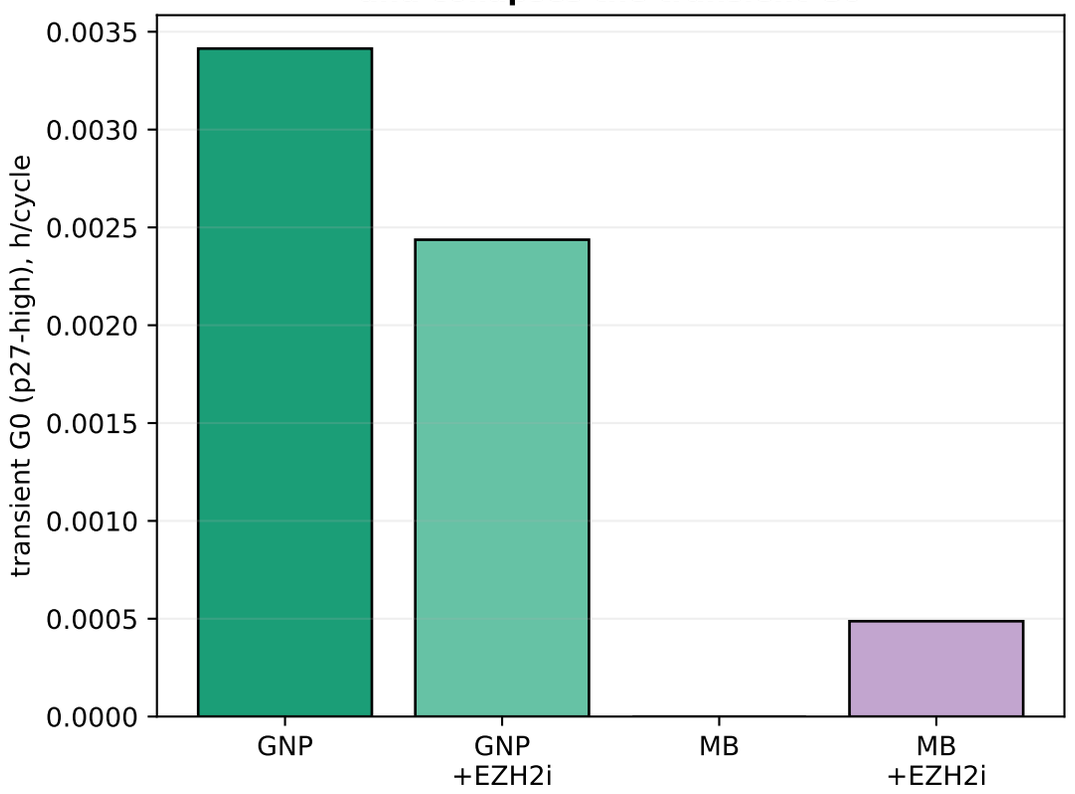

(a) CyclinD1 sets transient-G0 duration
(all knobs collapse; arrest below threshold)
SHH dose
translation
EZH2 repression
MB
transient G0 (p27-high), h/cycle
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0
2
4
6
8
10
12
14
16
CyclinD1 protein (mean, a.u.)


(b) Phase partition vs CyclinD1
(G0 shrinks into G1/S; period ~constant)
G0
G1
S
G2
phase duration (h)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
2.08
2.36
2.61
2.86
3.10
3.57
CyclinD1 protein (a.u.)


(c) One cycle (GNP): the transient G0 is p27-high / E2f-low,
NOT marked by phospho-Rb
transient G0
(p27-high, ~0 h)
CyclinD1 (norm)
phospho-Rb (norm; saturates early!)
p27 (=P21)
E2f (norm)
replication aRc (norm)
level (norm / a.u.)
0.0
0.2
0.4
0.6
0.8
1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
time since division (h)


(d) EZH2 inhibition de-represses CyclinD1
and collapses the transient G0
transient G0 (p27-high), h/cycle
0.0000
0.0005
0.0010
0.0015
0.0020
0.0025
0.0030
0.0035
GNP
GNP
+EZH2i
MB
MB
+EZH2i